

Advantages and Disadvantages of Polymer Shaping Methods

Method	Advantages	Disadvantages
Injection Moulding	High precision and repeatability Fast cycle times for mass production Good surface finish	High tooling cost Limited to thermoplastics Difficult for very large parts
Gas-Assisted Injection Moulding	Hollow parts with less material Reduces warping and sink marks Good for complex shapes	More complex tooling Process control is critical Only for thermoplastics
Extrusion	Continuous process Excellent for long, uniform profiles Low material waste	Limited to simple cross-sections Surface finish may need post-processing
Blown Film Extrusion	Produces thin, biaxially oriented films High throughput Ideal for packaging	Limited to thin film products Requires precise control
Blow Moulding	Great for hollow containers (e.g., bottles) Low cost for large volumes Good for thin-walled parts	Poor control over wall thickness Limited to simple shapes
Vacuum Forming	Low tooling costs Fast setup for low-volume production Simple equipment	Limited to shallow parts Poor detail resolution Only one good surface
Rotational Moulding	Uniform wall thickness Ideal for large hollow items No seams or welds	Long cycle time Lower detail accuracy Not suitable for small intricate parts
Compression Moulding	Good for thermosets Low scrap rate Can mould large parts	Limited to simple geometries Slower than injection moulding Labour intensive